A Verification Method for Game Image Assets Using C2PA Manifest Validation and Perceptual Hash-Based Similarity Analysis

Young-Sang Lee[1][0009−0009−0067−4009], Jung-Min Park[1][0009−0001−6751−5020], Won-Yong Lee[1][0009−0002−2516−2199], Geon Park[1][0009−0003−0417−1704], Myeong-Jae Lee[1][0009−0000−2419−8590], Yun-Ho Kim[1][0009−0007−8864−9611], Jung-Guen Lee[1][0009−0007−5614−028X], Dang Dai Tho[2][0000−0001−9327−1288], and Hye-Young Kim[3][0000−0001−7013−7510] ⋆

[1] Department of Games, Graduate School, Hongik University, Sejong, Republic of Korea
lys020207@naver.com, bjmpark12@naver.com, night98200@gmail.com, qkrrjs0773@gmail.com, audwolego@naver.com, kenshin2044@gmail.com, likename2@naver.com

[2] Department of Science, Technology and International Cooperation, Vietnam-Korea University of Information and Communication Technology, The University of Danang, Da Nang, Vietnam
ddtho@vku.udn.vn

[3] Department of Game Software, School of Games, Hongik University, Sejong, Republic of Korea
hykim@hongik.ac.kr

Abstract. Game image assets are repeatedly re-saved or transformed during collaboration, portfolio publication, web upload, and messenger-based sharing. This study evaluates a two-stage originality verification method that combines C2PA Manifest validation with perceptual hash-based similarity analysis. C2PA Manifests were embedded into 50 game image assets, and 850 transformed images and 250 control images were generated. Validation of 1150 files showed that Manifests were preserved in all signed originals and byte-level copies but were lost after ordinary re-saving and transformation. For files without a usable claim, aHash, dHash, pHash, and wHash accurately re-identified original candidates under common compression, resizing, color adjustment, format conversion, metadata removal, and watermarking. Performance decreased for screenshots, substantial cropping, and rotation. The results support using C2PA as primary evidence and perceptual hashing as supplementary evidence after provenance loss.

Keywords: C2PA · Manifest validation · Perceptual hash · Originality verification · Game image assets

⋆ Corresponding author.

1 Introduction

Game development projects produce character illustrations, backgrounds, UI icons, textures, and concept art that pass through graphics tools, repositories, messengers, and portfolio pages. Compression, resizing, cropping, format conversion, screenshots, and rotation can change a file while preserving part or all of its visual content. Consequently, it may be difficult to determine whether an image is an original file or a derivative of a known asset.

C2PA represents content provenance through Manifests, Claims, Assertions, and cryptographic signatures [1, 2]. A valid Manifest can provide strong file-level evidence, but re-encoding or metadata processing may remove the claim. Prior work has therefore noted the practical limits of provenance-only verification and the need for complementary media-authentication mechanisms [3–5]. Perceptual hashes offer such a complement because visually similar images can remain close in hash space even when their bytes differ [6–9].

This study evaluates whether C2PA survives realistic game-asset processing and whether aHash, dHash, pHash, and wHash can recover the correct original candidate after C2PA loss. It also identifies transformation conditions for which conventional whole-image hashing is insufficient.

2 Methodology

2.1 Dataset and Transformations

The dataset comprises 50 researcher-created game image assets: 10 backgrounds, 10 character illustrations, 10 concept-art images, 10 icons, and 10 textures. Commercial images and assets with uncertain copyright status were excluded. A C2PA Manifest containing the asset identifier, type, creator, generation tool, and basic provenance data was embedded into every original.

Five control conditions were applied to each original: byte copy, Pillow PNG resave, PNG metadata removal, JPEG Q95 conversion, and WebP Q90 conversion. Seventeen transformation conditions generated 850 additional queries, including JPEG compression, resizing, cropping, brightness and saturation adjustment, format conversion, metadata removal, watermarking, screenshot capture, and rotation. Together with the 50 signed originals, C2PA validation covered 1150 files.

2.2 Verification Procedure

C2PA validation recorded Manifest presence, signature validity, and data-hash validity. Because the experiment used a test certificate, a trust warning alone did not indicate failure; a file was counted as valid when its claim, signature, and data hash were valid.

For files without a usable claim, 64-bit aHash, dHash, pHash, and wHash values were compared with the 50-original database. For query q and candidate

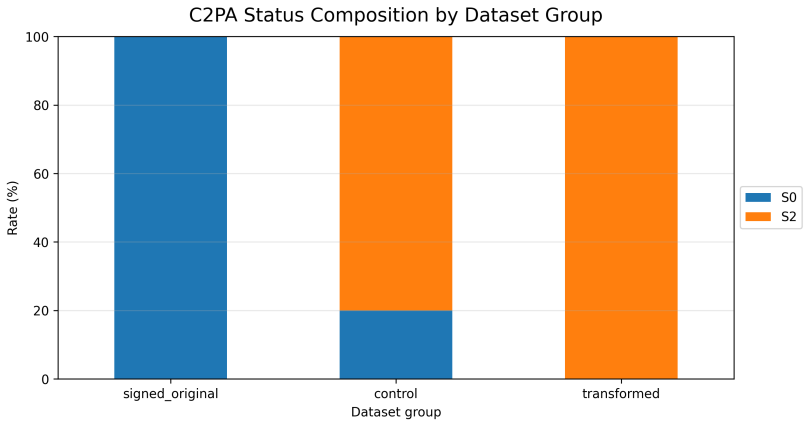


Fig. 1. C2PA status distribution by dataset group. Signed originals and byte copies remain verifiable, whereas re-saved and transformed files are predominantly in a no-claim state.

x_i, the predicted original was

$$x^* = \arg\min_{x_i} D_H\big(h(q), h(x_i)\big), \tag{1}$$

where D_H is Hamming distance. Top-1 accuracy measured whether x^* was the true original; Top-3 accuracy measured whether the true original appeared among the three nearest candidates.

3 Experimental Results

3.1 C2PA Manifest Preservation

The signed originals and byte-level copies retained detectable, valid C2PA claims. In contrast, re-saved control files and the transformed group returned no detectable claim. Figure 1 shows the resulting status composition. The signing procedure was reliable, but common image-processing pipelines removed file-level provenance.

3.2 Perceptual-Hash Identification

Figure 2 reports Top-1 accuracy by transformation condition. All four hashes performed strongly when the global visual structure was preserved. Byte copying, JPEG compression, resizing, brightness or saturation adjustment, format conversion, metadata removal, and watermarking were generally robust. All methods reached 100% under JPEG Q95, 50% resizing, metadata removal, and watermarking.

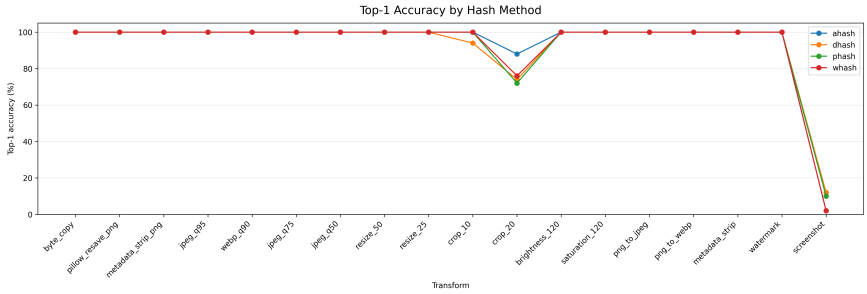


Fig. 2. Top-1 accuracy by perceptual-hash method and transformation condition.

Cropping, screenshots, and rotation were more difficult. Under 20% cropping, Top-1 accuracy ranged from 74% to 88%, while screenshot accuracy ranged from 0% to 8%. pHash achieved 82% under a 5-degree rotation, but no method exceeded 12% under 15-degree and 30-degree rotations. These cases should remain unknown or be subjected to additional review rather than being assigned automatically.

4 Conclusion

This study evaluated C2PA validation and perceptual hashing for originality verification of game image assets. C2PA provided strong evidence for signed originals and byte-level copies, but ordinary re-saving and transformation removed its claim. Perceptual hashing recovered original candidates under many visually preserving transformations, although screenshots, substantial cropping, and large rotations caused failure. A practical verifier should therefore use C2PA as primary evidence, apply perceptual hashing as supplementary evidence after claim loss, and defer structurally difficult or ambiguous cases for further review.

Acknowledgments. This research was supported by the Culture, Sports and Tourism R&D Program through the Korea Creative Content Agency grant funded by the Ministry of Culture, Sports and Tourism in 2024 (Project Name: Global Talent Training Program for Copyright Management Technology in Game Contents, Project Number: RS-2024-00396709, Contribution Rate: 100%).

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Coalition for Content Provenance and Authenticity: C2PA Technical Specification, Version 2.4. `https://spec.c2pa.org/specifications/specifications/2.4/specs/C2PA_Specification.html`, last accessed 2026/06/09
2. Coalition for Content Provenance and Authenticity: C2PA: Verifying Media Content Sources. `https://c2pa.org/`, last accessed 2026/06/09

3. Golaszewski, E., Krawetz, N., Sherman, A.T., Zieglar, E., Matukumalli, S.K., Yus, R., Kegley, C.L., Barthel, M., Bowman, W., Barot, B., Kullman, K.: Verifying Provenance of Digital Media: Why the C2PA Specifications Fall Short. arXiv preprint arXiv:2604.24890 (2026)
4. Simmons, J.C., Winograd, J.M.: Interoperable Provenance Authentication of Broadcast Media using Open Standards-based Metadata, Watermarking and Cryptography. arXiv preprint arXiv:2405.12336 (2024)
5. England, P., Malvar, H.S., Horvitz, E., Stokes, J.W., Fournet, C., Burke-Aguero, R., Chamayou, A., Clebsch, S., Costa, M., Deutscher, J., Erfani, S., Gaylor, M., Jenks, A., Kane, K., Redmiles, E., Shamis, A., Sharma, I., Wenker, S., Zaman, A.: AMP: Authentication of Media via Provenance. arXiv preprint arXiv:2001.07886 (2020)
6. Monga, V., Evans, B.L.: Perceptual Image Hashing via Feature Points: Performance Evaluation and Tradeoffs. IEEE Transactions on Image Processing **15**(11), 3452–3465 (2006). https://doi.org/10.1109/TIP.2006.881948
7. Swaminathan, A., Mao, Y., Wu, M.: Robust and Secure Image Hashing. IEEE Transactions on Information Forensics and Security **1**(2), 215–230 (2006). https://doi.org/10.1109/TIFS.2006.873601
8. Du, L., Ho, A.T.S., Cong, R.: Perceptual Hashing for Image Authentication: A Survey. Signal Processing: Image Communication **81**, 115713 (2020). https://doi.org/10.1016/j.image.2019.115713
9. Ou, Y., Rhee, K.H.: A Survey on Image Hashing for Image Authentication. IEICE Transactions on Information and Systems **E93-D**(5), 1020–1030 (2010). https://doi.org/10.1587/transinf.E93.D.1020